SYNTHESIS AND CHARACTERISATION OF Cu(II) AND Ni(II) COMPLEXES WITH TRIDENTATE N,O,N DONAR SHIFF BASE AND INTERACTION WITH HUMAN SERUM ALBUMIN (HSA) WITH Cu (II) COMPLEX

Project Report Submitted Towards the Practical Fulfillment

of the M.Sc degree

(3rd Semister in Inorganic Chemistry)

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❖ ACKNOWLEDGEMENT

With the best sincere effort and the best of my knowledge,I have prepared this project based on : “SYNTHESIS AND CHARACTERISATION OF Cu(II) AND Ni(II) complexes with TRIDENTED N,O,N-DONAR Schiff base : Interaction with human serum albumin(HSA) with Cu(II) complex.”

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CONTENT

❖ SYNTHESIS AND CHARACTERISATION OF Cu(II) AND Ni(II) COMPLEXES WITH TRIDENTED N,O,N DONAR SHIFF BASE: INTERACTION OF HUMAN SERUM ALBUMIN WITH Cu (II) COMPLEX

• ABSTRACT

A tetra coordinated copper (II) and nickel (II) complexes formulated as [Cu(L)(Cl)] (**1**) and [Ni(L)(Cl))] (**2**) were synthesized and characterized by elemental, physico-chemical and spectroscopic methods and X-Ray diffraction study. The four coordination spheres of metal (II) ions have been satisfied by one imine nitrogen- N and one phenolate oxygen-O and of organic moiety (**HL**).The interactions of copper (II) complex towards human serum albumin (HSA) was examined with the help of absorption spectroscopic tools. These study shows that the ligand behaves as a tridented ligand.

Key words: Transition metal complexes; Schiff base; Binding with HSA, Electrochemical study.

- # INTRODUCTION

Metal complexes of Schiff base are studied extensively due to synthetic flexibility of their compounds and there selectivity as well as sensitivity towards the central metal atom, i.e. versatile nature. Schiff bases contain N O N organic moiety have played an important role in the development of coordination chemistry as they readily formed stable complexes with most of the transition metals. The chemistry Schiff base ligands species has been gaining considerable interest primarily because of their functional structural derivatives. [1-7]. Schiff bases and their coordinating compounds are well known to be biologically important of interest for their antibacterial, antifungal, anti-inflammatory, antitubercular, anticancer, antineoplastic, anti-ischemic, anti-allergic, antihypertensive and antiulcerative agents [1-5].

As well as analytical reagent [16-18], fluorescent agents, polymer- coating, ink, pigment [19] Fluorescent materials [20] and catalytic reagents [21]. However changes in the electronic, steric and geometric properties of the legands alter the orbital at the metal centre and thus effect its properties. Many of them exhibit ability to bind toxic and heavy metals[12],undergo tautomerism[13],catalytic reduction [14] and photochromism [15]. Some of them have been used as complexing agents and powerful corrosion inhibitors. Schiff bases belong to a widely used group of organic intermediates important for production of special chemicals, e.g. pharmaceuticals, or rubber additives and as amino protective groups in organic synthesis. They also have uses as liquid crystals and in analytical, medicinal and polymer chemistry. Conventionally Schiff bases have been prepared by refluxing mixtures of the amine and the carbonyl compound in an organic solvent, for example, ethanol or methanol, but variations are known. In general, ketones react more slowly than aldehydes and higher temperatures and longer reaction times are often required as a result. Based on these facts, we decided to synthesize Schiff base of 2-(2-aminophenyl)-1-H-benzimidazole and 5-nitro salicyldehyde in ethanol medium using acid catalyst and the products were isolated simply by filtration. Here we report an account the synthesis, structural characterization and protein binding study of synthesized new ligand and their complex with Cu(II) and Ni(II) ions respectively in this paper.

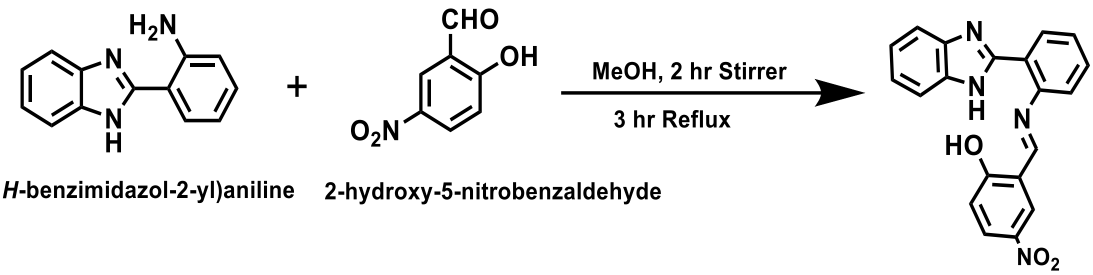
EXPERIMENTAL

➢ MATERIALS AND PHYSICAL MEASUREMENTS :

All chemicals and reagents were obtained from commercial sources and used as received, unless otherwise stated. Solvent (MeOH , DMF) were distilled from a appropriate drying agent. The complexes were synthesized by using the following procedure. Electronic absorbtion spectra was recorded on a SHIMADZU UV-1800 spectrometer. Cyclic voltametetry study have been examined.

➢ PREPARATION OF LIGAND

The ligand **HL** was Synthesize by adding 1.045 g (5.0 mmol) of 2-(1H-benzimidazol-2-yl)anailine and then 0.835 g (5.0 mmol) of 5-nitro salicyldehyde with 10 mL of ethanol in a round bottom flask with magnetic stirring given- in **Scheme 1**. Stir the solution for 2 hour and then reflux up to 3 hour and kept overnight to get the precipitate of the saffron solid ligand. The precipitate was filtered by using vacuum pump and washed several times using ethanol to remove any unreacted materials, finally crude product was collected by recrystallization in ethanol and dried the solid product as much as possible, Yields >80%.



2-(1*H*-benzimidazol-2-yl)aniline **2-hydroxy-5-nitrobenzaldehyde**

2-(((2-(1*H*-benzimidazol-2-yl)phenyl)imino)methyl)-4-nitrophenol

Scheme 1: synthetic strategy of ligsand HL fig. I.

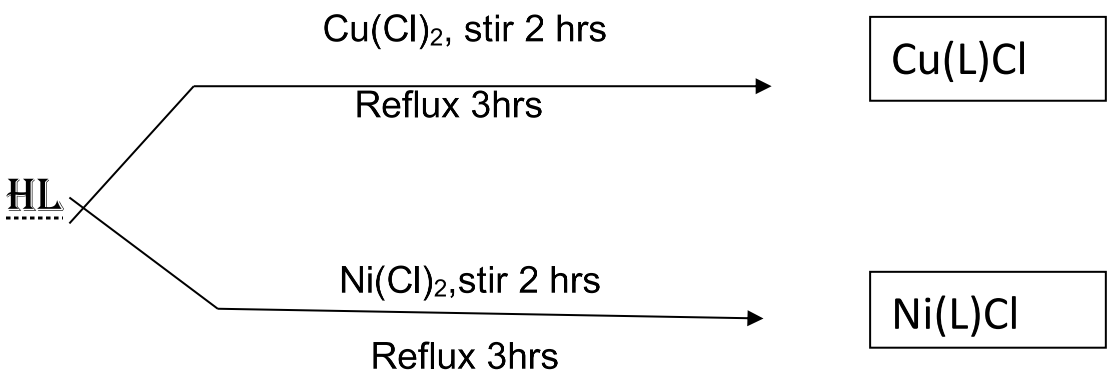
The ligand, was obtained from commercial sources with the known physical properties. The structure of the ligand was given above in fig 1:

➢ PREPARATION OF Ni(II)- LIGAND COMPLEX :

To prepare Nickel(II) complex, a common procedure was followed described below, using the nickel chloride salt and the organic ligand (HL) in the molar ratio 1:1 respectly. For this 0.0895gm(1/4mmol) of the ligand was dissolved in a round bottom flask by DMF and methanolic solution of 0.0324gm(1/4mmol) of the nickel(II) chloride was added dropwise into the ligand with constant stirring condition for 2 hours followed by refluxing 3 hours. The crude product was collected from the vacuum filtration in a crucible and then washed with methanol and kept aside to dry. The whole reaction was done in the methanolic medium.

➢ PREPARATION OF Cu (II)- LIGAND COMPLEXES :

To prepare copper (II) complex the reactants were properly weighed such that the ratio of ligand and metal became 1:1 respectively. For this 0.0716gm (1/5 mmol) of ligand was dissolved in a round bottom flask by DMF and methanolic solution of 0.0341gm (1/5 mmol) of the copper(II) chloride was addeddropwise into the ligand with constant stirring condition for 2 hours , followed by refluxing the mixture for 3 hours. The crude product was collected by vacuum filtration in a crucible, washed with methanol and kept aside to dry. The whole reaction is done in methanolic medium.



Scheme 2 : synthetic strategy of the cu(II) and ni (II) complexes

• RESULT AND DISCUSSION

The direct reaction of the ligand with Cu(II) and Mn(II) ions in methanolic medium using 1:1 ligand to metal ratio afford complexes of the type [ML(Cl)], where M = Cu(II) and Ni(II). In these complexes the ligand acts as a tridented chelating ligand through N,O,N-donar sites. These complexes are air stable, coloured solid, amorphous and soluble in DMF. The element analysis data of the ligand and their complexes (given in the table 1) are consistent with the calculated results from the empirical formula of each compound. The probable structures of the complexes is given in fig.1.

TABLE-1 : ELEMENTAL (C H N) ANALYSIS :

Formula	Compound	Molecular Weight	Colour	Calculated percentage				M.P. (°C)
				C(%)	H(%)	N(%)	M(%)	
$C_{20}H_{14}N_4O_3$	HL	358	Saffron	67.03	3.91	15.64	_	167
$C_{20}H_{12}N_4O_3Cu$	[CuL(Cl)]	419.5	Black	57.21	2.86	13.34	15.13	>250
$C_{20}H_{12}N_4O_3Ni$	[NiL(Cl)]	414.6	Yellow	57.88	2.89	13.50	14.13	>250

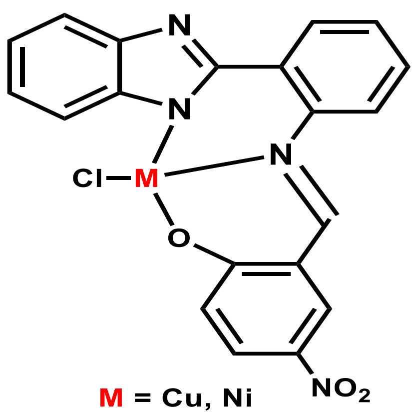


Fig.2: Probable structure of metal ligand complexes ; where m= Cu & Ni

• ELECTRONIC ABSORPTION SPECTRAL STUDY

The electronic spectra of the ligand and its complexes were recorded in methanol at room temperature **(Fig.3)**. The spectra of the Schiff base HL exhibit two main peaks: at 398.0 nm and 280.0 nm. The first and second peaks were attributed to benzene $n \rightarrow \pi^*$ and imino $\pi \rightarrow \pi^*$ transitions. All the spectra of complexes shows lower bands than 400 nm are due to intramolecular $\pi \rightarrow \pi^*$ and $n \rightarrow \pi^*$ transitions for the aromatic ring.

The Ni (II) complex, the spin allowed transition are expected from the energy level diagram for d^8 system due to $A_2 \rightarrow T_2$, $A_2 \rightarrow T_1(P)$ transition which are observed at low to high wavelength 283nm and 303 nm respectively.

Copper (II) complex shows peak at 275 nm and 395 nm. 275 and 395 nm peaks are attributed to intramolecular $\pi \rightarrow \pi^*$ and $n \rightarrow \pi^*$ transitions for the aromatic ring.

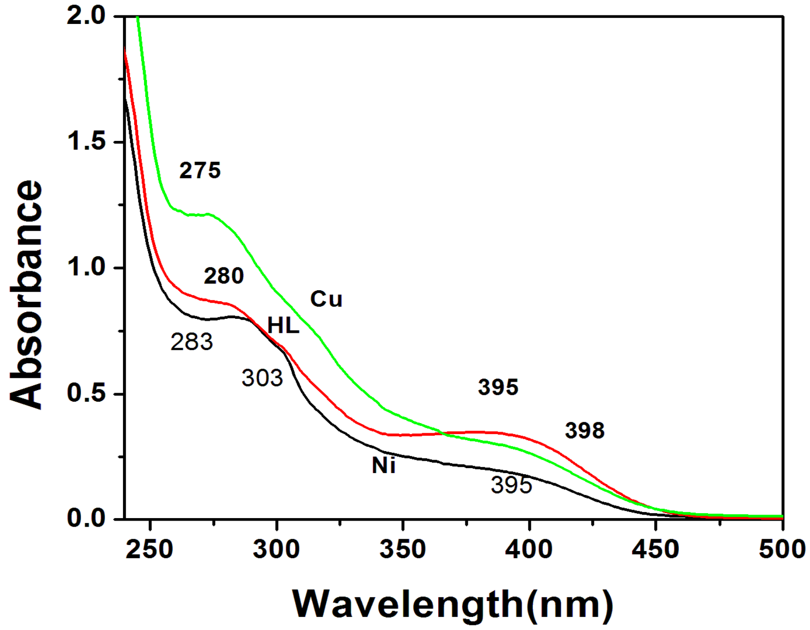


Fig. 3 : UV-VIS spectra of ligand and corresponding Cu(II) and Ni(II) complexes

• CV STUDY

The redox properties of the copper (II) complex was examined by cyclic voltammetry using a Pt-disk working electrode and a Pt-wire auxiliary electrode in dry dimethylformamide using [n-Bu_4N]ClO_4(0.1 M) as the supporting electrolyte. The cyclic voltammograms exhibit quasi-reversible transfer process with a reduction peak at E_{pc}= -0.32 V and with a corresponding oxidation peak at E_{pa}= 0.82 V for complex at a scan rateinterval 50–400 mV s^{-1}. The ratio of cathodic to anodic peak height was close to one. From these data, it can be deduced that the redox couple is related to a quasi-reversible one-electron transfer process controlled by diffusion

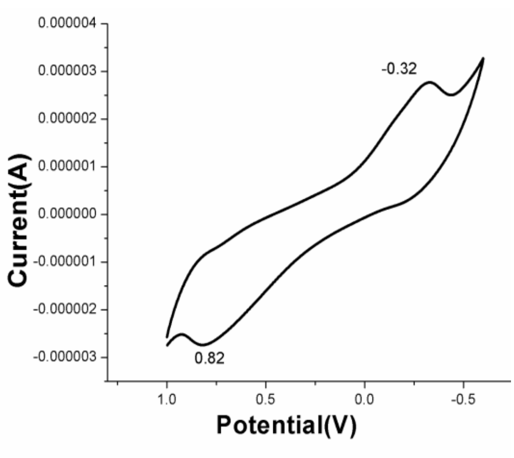
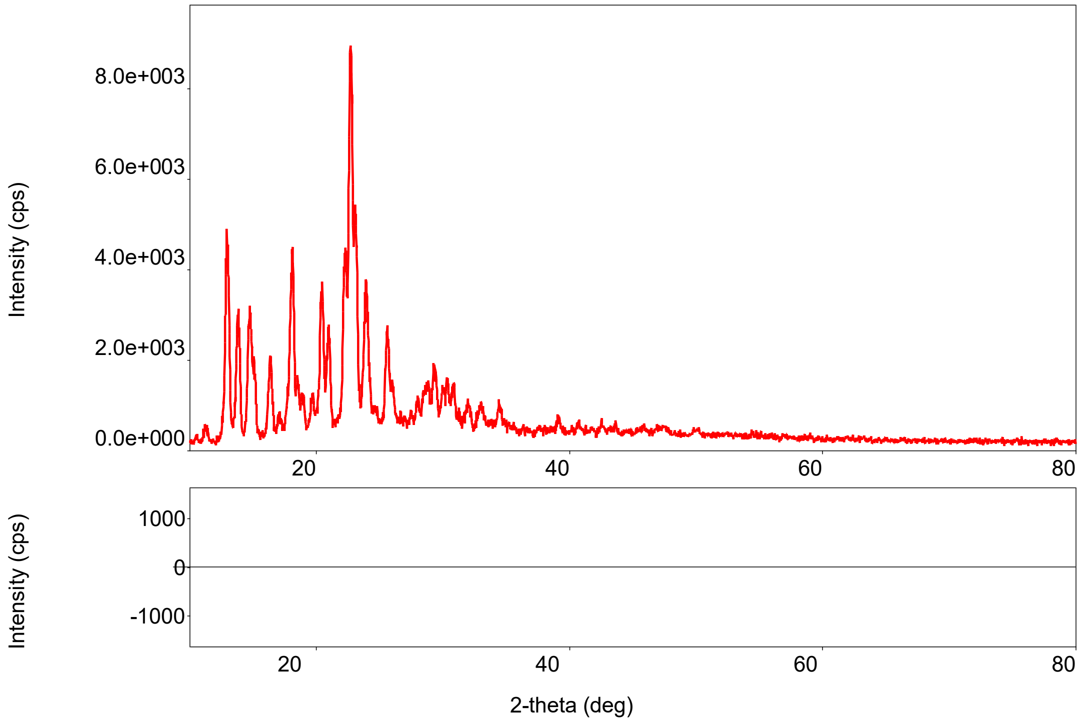


Fig.4 : Quasi-reversible transfer process in CV study of Cu(II) complex

- **X-RAY DIFFARCTION STUDY**



The sharp peak in X-ray diffraction study suggests that the compound is pure crystalline solid , it also helps to calculate h,k,l values.

- ## ABSORPTION CHARACTERISTICS OF HSA –CU(II) COMPLEX:

The absorption spectra of HSA in the absence and presence of Cu(II) complex was studied at different concentrations. From this study we observed that absorption of increases regularly upon increasing the concentration of the complex. The absorption spectral data of copper complex with HSA shown in **Fig.5**. It is may be due to the adsorption of HSA on the surface of the complex [15]. From these data the apparent association constant (K_{app}) determined of the complex with HSA has been determined using the **Benesi-Hildebrand equation[16].**

$1/(A_{obs} – A_0) = 1/(A_c – A_0) + 1/ K_{app}(A_c – A_0)[comp]$

Where, A_{obs} is the observed absorbance of the solution containing different concentrations of the complex at 280 nm, A_0 and Ac are the absorbances of HSA and the complex at 280 nm, respectively, with a concentration of complex and K_{app} represents the apparent association constant. The enhancement of absorbance at 280 nm was due to adsorption of the surface complex, based on the linear relationship between $1/(A_{obs}- A_0)$ *vs* reciprocal concentration of the complex with a slope equal to $1/K_{app}(A_c - A_0)$ and an intercept equal to $1/(A_c - A_0)$. The value of the apparent association constant (K_{app}) of HSA is 4.28×10^{-4} (R = 0.99897) determined from this plot (**Fig.6)** and represent a good linear relationship [17].

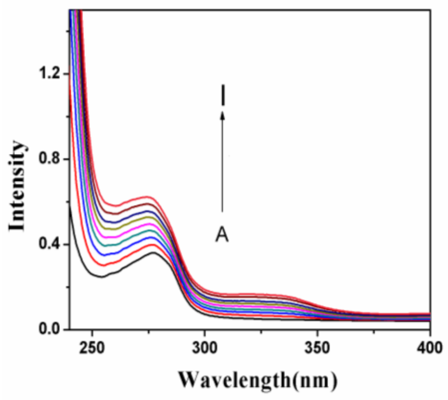


Fig. 5.

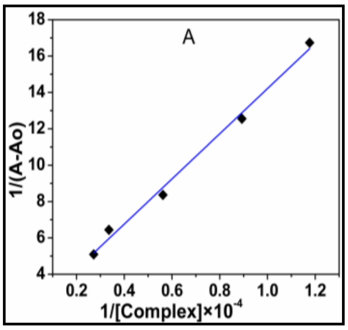


Fig.6.

Absorbtion spectra of HSA with Cu(II) complex

- # CONCLUSION

The synthesis and characterization of two mononuclear Ni (II), Cu (II) and complexes with a N,O,N- donor set have been performed. The ligand H_2L behaves as a N,O,N-donors.Melting point experiments show that the synthesized Schiff base complexes exhibit a significant thermal stability. On the basis of the physical and spectral data of the Schiff base and the complexes discussed above, one can assume that the metal ions are bonded to the Schiff base via the phenolic oxygen and the imino nitrogen ,all the complexes are tetrahedral geometry.The interactions of copper (II) complexes towards HSA were examined with the absorption spectroscopic tools. The absorption spectral titration indicated that Cu (II) ion strongly bind with HSA protein.The two complexes are characterized by UV–Vis and electrochemical techniques.

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